

QUANTIZATION CHANGES THE DATA: CLOSED-LOOP-AWARE QUANTIZATION FOR VISION-LANGUAGE-ACTION POLICIES

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ABSTRACT

Post-training quantization (PTQ) of a vision-language-action (VLA) policy must preserve task success, not just tensor fidelity. Unlike a predictor evaluated on fixed inputs, a compressed policy can change its future observations through its actions. This feedback motivates evaluating complete precision configurations and conditioning activation quantization on realized inputs. We introduce DYPAC-VLA, a training-free framework combining exact-byte precision adaptation with input-conditioned 8-bit activation ranges. An action-weighted representation-stability allocator supplies the initial mask; a complete-policy deviation audit enforces conservative acceptance of refinements. Across 50 RoboCasa365 tasks and 2,500 episodes per headline configuration, GR00T reaches **54.0%** success at $2.22\times$ static compression, compared with 30.4% for size-matched QuantVLA and 55.1% for FP16. On $\pi_{0.5}$, adapting an over-budget initializer yields 27.7% success at $2.70\times$ compression, versus 26.2% for FP16. A matched GR00T intervention isolates activation-range construction: input-conditioned ranges obtain 269/500 successes versus 0/500 for the registered frozen-range control. A compression sweep and a 4,000-episode LIBERO comparison establish both transfer and limits: more FP16 layers need not yield higher success, and deviation-based acceptance does not establish success optimality. These results identify deployment quantization as a consequential part of closed-loop policy compression, while separating storage savings from runtime acceleration.

1 INTRODUCTION

Vision-language-action (VLA) models map visual observations and language instructions to robot actions [Brohan et al. \(2023\)](#); [Octo Model Team et al. \(2024\)](#); [Kim et al. \(2024\)](#); [Black et al. \(2024\)](#); [Physical Intelligence et al. \(2025\)](#); [NVIDIA et al. \(2025\)](#). Their multimodal backbones and iterative action generators make model storage a practical deployment constraint. Post-training quantization (PTQ) reduces this footprint without retraining [Xiao et al. \(2023\)](#); [Lin et al. \(2024a\)](#). For a robot, however, the useful outcome is a compact policy that still completes its tasks. A small reconstruction error on calibration tensors is an intermediate criterion, not that deployment objective.

Closed-loop control makes this distinction consequential. A quantized action can change the next observation, which changes the activations and errors encountered by the policy thereafter. Calibration on a fixed teacher buffer does not directly evaluate this policy-dependent input distribution. The practical discrepancy can be large: on 50 RoboCasa365 tasks, two GR00T configurations with approximately 0.90 GiB of static storage obtain 30.4% success for QuantVLA and 54.0% for our deployment, compared with 55.1% for the larger FP16 reference. Similar storage therefore does not imply similar behavior. This motivates our central question: *how should VLA quantization account for the behavior and inputs induced by the compressed policy?*

We study this question through complete-policy deviation and deployment-conditioned activation ranges. Ordered action errors expose accumulated bias and replanning discontinuities that isolated tensor errors omit; complete precision configurations expose interactions that additive layer scores approximate; and input-conditioned ranges avoid relying exclusively on frozen calibration coverage. We refer to the missing dependence on execution context as the *full-context gap*. Closed-loop

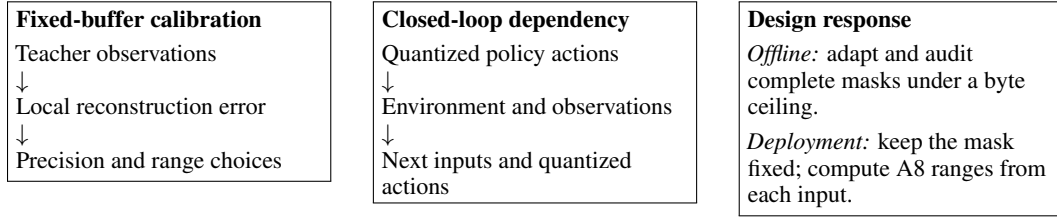


Figure 1: Why closed-loop policy compression differs from fixed-data calibration. The center depicts a feedback dependency: each action can affect the inputs to later decisions. The right separates offline precision adaptation from input-conditioned deployment. This dependency motivates the method; matched experiments test deployment choices without isolating every source of state-distribution shift.

error propagation itself is not new, and these considerations do not amount to solving the induced-distribution problem. Our contribution is to connect them to concrete VLA quantization choices and test their behavioral consequences with matched interventions.

We introduce DyPAC-VLA, a training-free framework with an offline precision stage and an input-conditioned deployment stage. Offline, a representation-stability allocator supplies an initial mixed W4/FP16 mask. Full-Context Precision Protection (FCPP) evaluates complete-policy interventions to fit a transferred mask within an exact byte ceiling and admits later refinements only under predeclared deviation guards. At deployment, a standard per-forward, per-channel max-abs rule computes 8-bit activation ranges from the actual input. The precision mask stays fixed: the policy neither routes by task nor applies output correction. The method combines established quantization primitives with an explicit behavioral audit; the evidence for each component is evaluated separately.

The results distinguish system effectiveness from component necessity. GR00T achieves 54.0% success at $2.22\times$ static compression, a 23.6-point improvement over size-matched QuantVLA. A fixed-mask intervention changes only activation-range construction and yields 269/500 successes with input-conditioned A8 versus 0/500 for the registered frozen table. On $\pi_{0.5}$, the precision stage converts an over-budget initializer into a feasible policy with 27.7% success at $2.70\times$ compression; FP16 obtains 26.2%. The latter difference is descriptive, rather than evidence that quantization generally improves policy capability. A six-arm GR00T sweep is non-monotonic in FP16 layer count, and the GR00T audit retains its initializer. These outcomes delimit what the framework establishes, while a 4,000-episode LIBERO matrix tests cross-benchmark transfer.

Our contributions are threefold.

- (1) **Closed-loop quantization analysis.** We formulate compression as task success under an exact storage budget and distinguish local reconstruction, complete-policy deviation, and realized-input range construction. Matched interventions test the consequences of deployment choices rather than assuming that offline fidelity predicts success.
- (2) **Budgeted adaptation and input-conditioned deployment.** We combine a guarded initializer with complete-policy precision adaptation and standard dynamic A8. On $\pi_{0.5}$, adaptation removes 41 FP16 protections to meet the target budget; on GR00T, the audit retains the feasible initializer without establishing its optimality.
- (3) **Success-storage evidence and boundaries.** We evaluate two architectures on RoboCasa365 and LIBERO, isolate post-allocation effects, and report compression sweeps and rejected candidates. The results support competitive compressed policies while exposing limits of proxy ranking, mixed-precision necessity, and runtime efficiency.

2 RELATED WORK

Efficient VLA deployment. Generalist VLA policies combine multimodal encoders with continuous-control heads Brohan et al. (2023); Octo Model Team et al. (2024); Kim et al. (2024); Black et al. (2024); Physical Intelligence et al. (2025); NVIDIA et al. (2025). Compact architectures, token reduction, temporal reuse, and architecture-quantizer co-design can reduce computation as well as storage Shukor et al. (2025); Zhang et al. (2025); Yang et al. (2025); Xu et al. (2025); Fang

et al. (2025). We retain the pretrained architecture and study its success–storage tradeoff. Runtime efficiency is a separate empirical question: Appendix A.12 reports both the memory benefit and latency cost of the evaluated backend.

Action-aware VLA quantization. QuantVLA adapts distribution transformations to language and diffusion-action stacks Zhang et al. (2026); ActQuant allocates tensor precision using action contribution and action-aware curvature Akbari et al. (2026); Ω -QVLA combines rotations with denoising-step scaling Wang et al. (2026b); and DA-PTQ uses trajectory motion drift for mixed-precision VLA quantization Xu et al. (2026a). These approaches already move beyond generic tensor reconstruction, so action awareness alone is not our distinguishing claim. We couple complete-mask deviation auditing under exact byte constraints with input-conditioned activation ranges, and use controlled rollouts to examine where these choices help. Local architecture adaptations of ActQuant and DA-PTQ are explicitly separated from their source protocols.

Calibration and dynamic activation ranges. GPTQ and AWQ use second-order information or activation salience for weight quantization Frantar et al. (2023); Lin et al. (2024b). Weight–activation PTQ uses smoothing, calibration, or rotations Xiao et al. (2023); Shao et al. (2024); Ashkboos et al. (2024); Lin et al. (2024a); Liu et al. (2025); Sun et al. (2025), while diffusion PTQ addresses timestep-dependent ranges and iterative error accumulation Shang et al. (2023); Li et al. (2023); Wu et al. (2024); Chen et al. (2025); Zhao et al. (2024). Input-dependent scaling is established in ZeroQuant Yao et al. (2022) and ViDiT-Q Zhao et al. (2025). Our per-channel max-abs estimator is likewise standard. The question here is its behavioral importance when a quantized policy acts in an environment, not a new scalar quantizer or a claim that all fixed calibration schemes fail.

Precision allocation and behavioral proxies. Vision–language PTQ studies compensation, modality-aware channels, token saliency, and cross-layer dependencies Deng & Huang (2026); Xue et al. (2026); Wang et al. (2024; 2026a). Mixed-precision allocation uses hardware feedback, Hessian sensitivity, reconstruction loss, or diffusion-specific proxies Wang et al. (2019); Dong et al. (2019; 2020); Hubara et al. (2021); Li et al. (2021); Zhao et al. (2024). We combine established geometry and distribution diagnostics Kornblith et al. (2019); Yin et al. (2026) with action-impact weights and byte costs to construct an initializer. Complete-policy scoring then audits proposals rather than treating additive layer benefits as sufficient. Our experiments distinguish this deviation-based decision rule from predicting task success: neither a rejected candidate nor a retained initializer establishes success optimality.

3 METHOD

Overview. DYPAC-VLA separates two decisions: which precision configuration fits the storage budget, and how that configuration quantizes inputs during execution. We first state the closed-loop evaluation objective, then describe the initializer and complete-policy deviation audit used for offline adaptation. Input-conditioned A8 supplies the deployment rule. The output is one frozen mask and one activation rule; neither success feedback nor task routing is available at runtime.

3.1 THE FULL-CONTEXT GAP UNDER AN EXACT STORAGE BUDGET

Let $f_\theta(o, \epsilon)$ denote a frozen FP16-reference VLA that maps observation o and generative noise ϵ to an action chunk $A \in \mathbb{R}^{H \times d}$. The target linear layers are indexed by $\mathcal{L}$. A mask $M \in \{4, 16\}^{|\mathcal{L}|}$ assigns signed-nibble W4 or native 16-bit precision to each target, and $Q_M(f_\theta)$ denotes the mixed-precision policy under a fixed activation rule. Here FP16 follows the reference-row naming; native tensors may use BF16 in the model adapter. Let $\mathcal{D}_M$ be the rollout distribution induced by this policy under a fixed task distribution, and let $S(Q_M; \tau) \in \{0, 1\}$ denote task success on rollout τ . The deployment objective is

$$M^* = \arg \max_{M \in \{4, 16\}^{|\mathcal{L}|}} \mathbb{E}_{\tau \sim \mathcal{D}_M} [S(Q_M; \tau)] \quad \text{s.t.} \quad C(M) \leq C_{\max}, \quad (1)$$

where $C(M)$ is exact packed static-component storage. Equation 1 states the evaluation goal, not the selection rule: success labels are unavailable during FCP, and collecting $\mathcal{D}_M$ for every mask is

intractable. We therefore screen complete policies with label-free physical-sequence and component-deviation guards on teacher-induced contexts $\mathcal{D}_T$, then collect candidate-induced contexts only for shortlisted masks. Section 3.3 constructs anchor M_0 , defines the finite audited class $\mathcal{C}(M_0)$, and returns $\widehat{M}$ with abstention.

The dependence of both success and fidelity on $\mathcal{D}_M$ follows from standard closed-loop error propagation. For a local linearization with action error $e_t = a_t^Q - a_t^T$,

$$\delta x_{t+1} = A_t \delta x_t + B_t e_t, \quad \delta x_T = \sum_{t=0}^{T-1} \Phi_{T,t+1} B_t e_t, \quad (2)$$

where $\Phi_{T,t+1}$ composes the intervening state Jacobians. Per-step MSE measures $\|e_t\|^2$ but ignores error propagation, directional coherence, and the observations reached after a deviation. This motivates a sequence-level contract and candidate-state audit; closed-loop rollouts remain the final measure of task success.

FCP is label-free: success and reward never enter its guards. Task identifiers only stratify uncertainty; candidate and reference share generative noise; and an independent noise stream is reserved for post-freeze audit. The upstream initializer is not label-free because a disclosed four-task success sweep chooses one objective weight. The final mask and activation rule are global. No gradient update, task-conditioned routing, output correction, or FP16 bypass is permitted.

Before deployment, FCP may use the FP16 teacher and unlabeled on-policy observations; during deployment, it cannot observe success or change precision. The mask is thus a fixed, auditable artifact rather than a runtime selector.

3.2 ACTION-WEIGHTED CS-CKA PRECISION ALLOCATION

The allocator asks where a compressed policy should spend its limited FP16 capacity. CKA measures the geometry retained by a layer intervention, whereas CS-Aligner measures its distribution shift Kornblith et al. (2019); Yin et al. (2026). For target i , we combine jointly normalized geometry loss $\bar{d}_i^g = \text{norm}(1 - \widehat{\text{CKA}}_i)$ and distribution shift $\bar{d}_i^d = \text{norm}(\widehat{D}_{\text{CS},i})$:

$$S_i = \frac{\lambda_g \bar{d}_i^g + \lambda_d \bar{d}_i^d}{\lambda_g + \lambda_d}, \quad (3)$$

$$M_{\text{init}} \approx \arg \min_M \sum_i \mathbf{1}[M_i = 4] w_i S_i \quad \text{s.t.} \quad C(M) \leq C_{\text{max}}, \quad M_i = 16 \quad (i \in \mathcal{G}).$$

Here w_i is a winsorized action-impact weight, $\mathbf{1}[M_i = 4]$ charges the estimated cost of quantizing target i , and $\mathcal{G}$ contains targets excluded from W4 by RMS and saturation guards. Thus, precision follows the action-weighted joint cost of representation geometry and distribution stability under the actual byte constraint. Native allocation can already satisfy the ceiling, whereas transferring its protection pattern to a different architecture can violate the target budget. FCP handles this latter case by using complete-policy evidence to decide which protections to remove. A four-task development sweep fixes $\lambda_g : \lambda_d = 16 : 1$ from seven choices. A greedy byte-constrained initialization followed by fixed-size flips approximates this objective (Appendix A.1). This produces a starting configuration, whose necessity and signal fusion are tested rather than assumed.

We estimate the allocation evidence on FP16 on-policy observations from 12 tasks, with four tasks from each of Atomic-Seen, Composite-Seen, and Composite-Unseen. Three seeds per task and four replans per sequence yield 36 sequences and 144 observations. A task-seed hash assigns early, middle, or late rollout windows. Task identifiers stratify the statistics, but Equation 3 returns one global mask.

3.3 FULL-CONTEXT POLICY AUDIT

Physical-sequence contract. D_{PAC} measures deviation from teacher actions for a chunked, replanning VLA policy. It compares ordered physical actions over a replan sequence through coherent bias, pose drift, replan discontinuities, gripper changes, and tail deviation. FCP uses this proxy to audit complete-policy interventions. It is neither a task-success estimator nor a guarantee of behavioral preservation.

Paired physical sequences. For each task–seed sequence, teacher and candidate share observations and initial action noise. We inverse-normalize the final chunks and retain all $H = 16$ actions executed before each replan, yielding an ordered sequence without overlapping forecasts. Errors use one robust coordinate scale estimated globally from FP16 data, so a candidate cannot alter its denominator. Five terms then measure local action error, normalized prefix drift, right-composed SE(3) pose drift, replan-boundary discontinuity, and gripper state/event timing. The prefix term preserves coherent bias, and the last three expose control errors hidden by coordinatewise MSE. Appendix A.2 gives all formulas and edge cases.

The five normalized components receive equal weight:

$$\begin{aligned} d_{\text{PAC}}^{(u)} &= \frac{1}{5} (d_{\text{local}} + d_{\text{prefix}} + d_{\text{pose}} + d_{\text{stitch}} + d_{\text{grip}}), \\ D_{\text{PAC}} &= \text{mean}_u d_{\text{PAC}}^{(u)} + \text{CVaR}_{0.9,u} \left(d_{\text{PAC}}^{(u)} \right). \end{aligned} \quad (4)$$

Here u indexes ordered task–seed sequences; the outer CVaR gives the worst 10% the same total weight as the mean. We retain D_{func} as a complementary final-action and kinematic diagnostic, not a second closed-loop loss.

Abstaining complete-policy audit. FCP treats mask adaptation as constrained decision-making rather than a second allocation heuristic. Complete-policy coordinate evidence retains or projects M_{init} to obtain a byte-feasible anchor M_0 ; paired evaluation of complete proposal masks then decides whether to admit a later refinement. Thus, abstention can still follow a consequential feasibility projection: it returns M_0 , which equals M_{init} only when the initializer was already feasible.

Two-stage protocol. The construction stage pairs complete-policy sequences on a stratified buffer, evaluates instantiated one-layer interventions, projects an over-budget seed into the feasible set, and generates a finite set of exact-byte proposals. The acceptance stage applies task-level one-standard-error bounds to D_{PAC} , D_{func} , pose, stitch, and gripper deviation. Any surviving proposal is checked again on candidate-induced observations before it can replace M_0 .

Evidence-guided feasibility projection. If $C(M_{\text{init}}) \leq C_{\text{max}}$, FCP sets $M_0 = M_{\text{init}}$. Otherwise, it instantiates each FP16-to-W4 removal as a complete policy and removes protections from least to most damaging until the mask is feasible. This label-free projection ensures that abstention always returns a deployable mask; only the subsequent audit may replace M_0 .

Coordinate counterfactuals. Starting from M_0 , FCP flips target precision states one at a time and, when scheduled, repeats the coordinate sweep around the preceding winner. Each intervention is instantiated as a mixed-precision network and compared with its complete-mask background. The resulting conservative benefits feed an exact sparse Pareto 0–1 dynamic program subject to

$$C(M) = C_{\text{fixed}} + \sum_{i \in \mathcal{L}} C_i(M_i) \leq 1.10 C_{\text{QuantVLA}}. \quad (5)$$

The constraint is expressed in bytes rather than layer count because layers differ in tensor and metadata size. The dynamic program only generates candidates: its additive counterfactual benefits cannot capture interactions among simultaneous layer changes.

The finite class $\mathcal{C}(M_0)$ contains M_0 , dynamic-program proposals, and five structured proposal types. Their protected sets are rebuilt from the layer scores, not restricted to one-layer neighbors of M_0 : the GR00T proposals retain only one to five FP16 layers versus 16 in M_0 . A positive conservative FP16 benefit means that retaining or restoring FP16 is valuable; for an already protected layer it does not mean that removing protection improves the policy. Additive scores measured around M_0 may extrapolate poorly to these multi-layer changes. Complete-policy evaluation checks the emitted class, not the entire feasible neighborhood (Appendix A.11).

Task-level minimax adjudication. For metric $q \in \{D_{\text{func}}, D_{\text{PAC}}\}$ and task cluster $g \in \{\text{all}, \text{Atomic}, \text{C-Seen}, \text{C-Unseen}\}$, let $\Delta_{q,t}(M)$ be the paired candidate-minus- M_0 task scalar. We define

$$U_{q,g}(M) = \frac{\text{mean}_{t \in g} \Delta_{q,t}(M) + \text{SE}_{t \in g}^{\text{jack}}[\Delta_{q,t}(M)]}{\text{mean}_{t \in g} q_t(M_{\text{base}})}, \quad J(M) = \max_{q,g} U_{q,g}(M). \quad (6)$$

Here $M_{\text{base}} = M_0$ is evaluated on the same context as the candidate. The jackknife makes the task, rather than an observation, the uncertainty unit. Eligibility requires $J(M) < 0$ and non-positive one-SE upper bounds for pose, stitch, and gripper; these conservative guards are not physical-safety certificates.

Candidate-state check and freeze. The protocol reserves candidate-induced observations over four to eight replans for shortlisted candidates and defines $J_{\text{final}} = \max(J_{\text{teacher}}, J_{\text{candidate}})$. In the implemented selection order, eligible teacher-state candidates are first filtered by a paired one-standard-error comparison against the best teacher-state objective; up to three then enter candidate-state adjudication. Let $\mathcal{E}(M_0)$ denote the remaining byte-feasible candidates with negative J_{final} and the required deviation guards. The frozen choice is

$$\widehat{M} = \begin{cases} \text{lexmin}_{M \in \mathcal{E}(M_0)}(C(M), N_{16}(M), \text{names}(M)), & \mathcal{E}(M_0) \neq \emptyset, \\ M_0, & \mathcal{E}(M_0) = \emptyset, \end{cases} \quad (7)$$

where N_{16} counts protected layers and names provide deterministic ordering. Storage breaks the tradeoff only after deviation eligibility: the rule does not exchange increased proxy error for fewer bytes. On GR00T, all proposals fail the teacher-state screen, so candidate-state adjudication is not exercised. Retaining M_0 certifies neither local optimality nor prevention of a task-success regression.

3.4 DEPLOYMENT CONTEXT: INPUT-CONDITIONED ACTIVATION QUANTIZATION

The deployment stage removes the assumption that a frozen range table covers later closed-loop inputs. Hessian-aware W4 provides the weight substrate, while A8 uses a standard per-forward max-abs estimator. This simple rule isolates the deployment policy—conditioning ranges on realized inputs—rather than proposing a new scalar estimator.

Weight quantization. For each W4 layer, 256 calibration observations form FP16 input second-order statistics $X^\top X$. A GPTQ-style sequential error-feedback pass selects clipping and quantizes signed weights in groups of 64 [Frantar et al. \(2023\)](#). Two signed four-bit codes are packed per byte; per-group scales and packing metadata are retained, but the FP16 weight tensor is not. Row rotation is the identity, so the reported method does not rely on an unreported learned transform.

Input-conditioned activation range. A calibrated static activation table can become mismatched as the observation, flow step, and policy-induced state change. Our static counterfactual uses independently calibrated 99.9th-percentile scales for each target layer, input channel, and applicable flow step. DYRANGE-A8 removes this lookup and recomputes a scale for every target linear layer, forward call, and input channel:

$$s_c = \max\left(\frac{\max |x_{\dots, c}|}{127}, 10^{-6}\right), \quad q_c = \text{clip}(\text{round}(x_c/s_c), -127, 127). \quad (8)$$

The rule is deterministic and uses the input dtype. It has no calibration lookup, flow-step selector, task selector, or candidate branch. Equation 8 is applied uniformly, and runtime output correction is disabled. Appendix A.12 reports hardware measurements under explicitly audited runtime contracts.

Exact storage scope. We report the byte sum of fixed model components, packed W4 codes, per-group scales and metadata, and native FP16 tensors. This scope excludes activations, CUDA workspaces, simulator state, and eager fake-quant caches. The projected $\pi_{0.5}$ anchor contains 121 W4 and 59 FP16 targets and occupies 1,634,828,288 bytes; the native GR00T plan contains 100 W4 and 16 FP16 targets and occupies 962,068,480 bytes. These values establish static compression only. A separate A40 experiment measures runtime quantities directly.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Models, tasks, and evaluation. We evaluate GR00T N1.5 and $\pi_{0.5}$ [NVIDIA et al. \(2025\)](#); [Physical Intelligence et al. \(2025\)](#) on RoboCasa365 v1.0.1 [Nasiriany et al. \(2026\)](#); 18 Atomic-Seen, 16

Table 1: RoboCasa365 success–compression operating points. On $\pi_{0.5}$, DYPAC-VLA reaches 27.7% success at $2.70\times$ compression, above FP16 (26.2%) and every displayed compressed baseline.

Configuration	Allocation	Atomic SR $\uparrow$	C-Seen SR $\uparrow$	C-Unseen SR $\uparrow$	All SR $\uparrow$	Size (GiB) $\downarrow$	Comp. $\uparrow$
$\pi_{0.5}$ Physical Intelligence et al. (2025) (<i>formal target split</i>)							
FP16	–	58.3	14.0	2.1	26.2	4.113	1.00 $\times$
QUANTVLA W4A8	180 W4	56.1	12.8	1.8	24.8	1.388	2.96 $\times$
Uniform W6	180 W6	56.6	10.9	2.1	24.5	1.902	2.16 $\times$
Ω -QVLA W4A4 [‡]	252 W4	49.6	10.4	1.0	21.5	1.307	3.27 $\times$
ActQuant 4.0 BPW/A16 [§]	Tensor IQ2–Q4	56.6	12.3	2.8	25.2	2.904	2.15 $\times$
DA-PTQ W4A8	87 W4/15 BF16	58.6	12.3	3.8	26.2	1.390	2.96 $\times$
DYPAC-VLA (Ours)[†]	121 W4	59.6	15.4	4.3	27.7	1.523	2.70 $\times$
DYPAC-VLA (Ours, max rate)[#]	180 W4	58.7	14.5	3.3	26.8	1.157	3.56 $\times$
GR00T N1.5 (<i>high-success control</i>)							
FP16	–	75.6	41.9	45.3	55.1	1.993	1.00 $\times$
QUANTVLA W4A8	116 W4	49.6	19.6	19.8	30.4	0.898	2.22 $\times$
Uniform W6	116 W6	68.4	42.9	41.8	51.7	1.109	1.80 $\times$
Ω -QVLA W4A4 [‡]	180 W4	60.1	25.9	27.5	38.7	0.599	3.33 $\times$
ActQuant 4.0 BPW/A16 [§]	Tensor IQ2–Q4	72.1	42.6	41.5	52.9	3.655	1.39 $\times$
DA-PTQ W4A8	55 W4/9 BF16	8.1	0.0	0.4	3.0	1.110	1.80 $\times$
DYPAC-VLA (Ours)[†]	100 W4	74.7	43.9	40.9	54.0	0.896	2.22 $\times$
DYPAC-VLA (Ours, max rate)[#]	116 W4	70.9	42.9	42.3	52.8	0.779	2.56 $\times$

C-Seen/C-Unseen denote Composite-Seen/Unseen; $\pi_{0.5}$ uses four flow steps. Registered paired tests are reported in the text. [†]Ours includes initializer-selection tasks; exact bytes are audited. [#]Ours (max rate): all-W4 profiles at formal 2,500-episode scale (GR00T rate-family endpoint; $\pi_{0.5}$ 180-layer W4 profile); descriptive rows, cross-run vs. the audited anchors (GR00T 319/288, exact McNemar $p = 0.223$; $\pi_{0.5}$ 182/159, exact McNemar $p = 0.233$); not registered tests. [‡] Ω -QVLA uses model/task-set-specific calibration. [§]ActQuant Akbari et al. (2026) uses local FP16-teacher proxy calibration (source-protocol-equivalent=false) and follows the pinned official GitHub implementation. Its size is a measured static pack; GR00T reports the mean across its three split checkpoints. ^{||}DA-PTQ Xu et al. (2026a) is a complete local adaptation (2,500/model; source-protocol-equivalent=false); Appendix 10 gives source results. Compression is static storage, not runtime memory or latency.

Composite-Seen, and 16 Composite-Unseen tasks. Headline configurations use 50 seeds per task, the official horizon, four flow steps, and shared task–seed–replan noise (2,500 episodes each). Focused interventions use the same tasks with seeds 0–9. LIBERO tests transfer to four additional suites.

Comparators and storage. FP16 is the reference; QuantVLA W4A8 Zhang et al. (2026), Uniform W6, Ω -QVLA W4A4 Wang et al. (2026b), and local ActQuant/DA-PTQ adaptations provide compressed comparisons. W4 and A8 denote four-bit weights and eight-bit activations. Static storage includes fixed components, packed weights, scales, and metadata, but excludes activations and runtime workspaces. GR00T occupies 962,068,480 bytes, or $0.998\times$ the QuantVLA footprint. The projected $\pi_{0.5}$ mask occupies 1,634,828,288 bytes and satisfies its registered ceiling; this is not an exactly size-matched comparison to every baseline.

Statistics and selection provenance. We report task/split-macro success and exact paired two-sided McNemar tests, with Holm correction within registered families. Inferential claims are restricted to those comparisons; non-significance does not establish equivalence or non-inferiority. A four-task development sweep selected the GR00T initializer weighting from seven choices, and these tasks remain in the headline. On the other 46 tasks, post-freeze success is 51.8/27.6/52.8% for ours/QuantVLA/FP16. Frozen manifests bind configurations, code, coverage, and hashes; incomplete or invalid cells are suppressed.

4.2 CAN COMPACT POLICIES RETAIN CLOSED-LOOP SUCCESS?

GR00T: a size-matched behavioral contrast. DYPAC-VLA reaches 54.0% at $2.22\times$ static compression, compared with 30.4% for essentially size-matched QuantVLA and 55.1% for FP16. The 23.6-point gain over QuantVLA is significant under the registered paired family (Holm $p < 10^{-4}$); the comparison with FP16 gives $p = 0.2929$. Thus the compressed policy approaches the observed

reference success rate, without an equivalence claim. Its precision mask is unchanged by FCP, so this system-level gain cannot be attributed to selector-induced mask improvement.

$\pi_{0.5}$: **cross-architecture budget adaptation.** The feasible policy reaches 27.7% at $2.702\times$ compression, compared with 26.2% for FP16 and 24.8% for QuantVLA. It has the highest observed success among the displayed compressed configurations, whose storage and calibration protocols differ. The 1.5-point margin over FP16 is descriptive, not evidence that compression generally improves capability. Section 4.5 isolates how the transferred precision configuration meets the budget.

Adapted-baseline scope. Local ActQuant reaches 52.9/25.2% on GR00T/ $\pi_{0.5}$ versus our 54.0/27.7% (Holm $p = 0.3102/0.0011$). Local DA-PTQ reaches 3.0/26.2% ($p = 0.0648$ on $\pi_{0.5}$). Both require architecture adapters and proxy calibration and are labeled source-protocol-nonequivalent. In particular, the low GR00T DA-PTQ result characterizes this adaptation, not the native method’s performance on its original benchmark.

4.3 DOES THE SUCCESS-STORAGE TRADEOFF TRANSFER?

Table 2: Cross-benchmark LIBERO transfer. Cells report four-suite mean success rate (%) and tightly packed static component size.

Configuration	$\pi_{0.5}$		GR00T N1.5	
	Avg. SR $\uparrow$	Size (GiB) $\downarrow$	Avg. SR $\uparrow$	Size (GiB) $\downarrow$
FP16	96.8	4.113	86.8	1.993
QUANTVLA W4A8	96.3	1.388	81.3	0.898
Uniform W6	96.8	1.902	70.5	1.109
Ω -QVLA W4A4 [†]	98.0	1.307	84.5	0.599
DyPAC-VLA (Ours)	97.3	1.788	93.8	0.948

The joint local gate covers 4,000 episodes across 40/40 exact-valid suite cells. [†]The displayed $\pi_{0.5}$ Ω -QVLA success value is experiment-owner attested. Ours size is the arithmetic mean of four suite-specific static plans. Appendix A.19 reports the full suite breakdown and protocol.

The complete 4,000-episode LIBERO matrix yields 97.3% mean success on $\pi_{0.5}$ at 1.788 GiB and 93.8% on GR00T at 0.948 GiB. On GR00T, ours has the highest displayed success; on $\pi_{0.5}$, Ω -QVLA achieves 98.0% at a smaller 1.307 GiB. Transfer therefore supports competitive compressed policies, not universal dominance. Appendix A.19 reports suite-level coverage and size scope.

4.4 WHICH DEPLOYMENT CHOICES CHANGE BEHAVIOR?

Post-allocation quantization at a fixed mask. GR00T permits a matched 100-W4/16-FP16 comparison. Replacing the initializer-stage DuQuant/static-A8 stack with Hessian-aware W4 and input-conditioned A8 raises success from 50.8% to 54.0% (370/289 paired wins/losses; Holm $p = 0.0036$). Precision placement is identical, so this identifies a combined post-allocation effect rather than a gain from mask search.

Activation-range construction. Holding the mask and runtime fixed, the registered frozen-range control obtains 0/500 successes versus 269/500 for input-conditioned A8. All frozen-range episodes complete without a runtime or numerical failure. Independently, the measured median scale drift across applicable action-head layers is 10.8%, 16.8%, and 24.9% at flow steps 2–4, with a worst layer-step value of 49.0% (Appendix A.10). The intervention establishes the behavioral importance of range construction in this configuration. It does not isolate policy-induced state shift from calibration coverage or flow-step effects, nor establish failure of all static calibration schemes.

Precision count is not a monotonic predictor. A preregistered six-arm GR00T sweep holds the Hessian-g64/input-conditioned-A8 stack and layer ordering fixed. Success ranges from 51.4% to 55.4% while static compression spans $1.166\times$ – $2.556\times$ across 3,000 episodes. The all-W4 endpoint ties the least-compressed arm at 55.4%. At 2,500-episode scale, the descriptive all-W4 endpoint reaches 52.8% versus the anchor’s 54.0% (Table 1). These results do not establish that mixed precision is universally necessary, or that every extra protected layer improves behavior.

4.5 WHAT DOES COMPLETE-POLICY ADAPTATION ESTABLISH?

Feasibility projection on $\pi_{0.5}$. The transferred initializer protects 100 of 180 targets and exceeds the byte ceiling by 205,587,047 bytes. FCP evaluates all 180 one-layer interventions in complete-policy context and converts 41 protected targets to W4, yielding a feasible 121-W4/59-FP16 anchor. Later proposals fail the guards, leaving this projected configuration deployed.

Appendix Table 9 reports the paired mask diagnostic.

The registered 500-key diagnostic gives 28.2% for the projected anchor versus 26.0% for the initializer ($p = 0.235$). The projection demonstrably enforces the budget; the observed success difference is positive but does not prove non-inferiority or an optimal choice of removals.

Retention on GR00T. The GR00T initializer already fits the budget. All six emitted proposals fail the teacher-state deviation screen, so FCP returns M_0 without candidate-state adjudication. These proposals retain one to five FP16 layers rather than the anchor’s 16; the “single-layer” proposal is a one-protected-layer configuration, not a one-flip neighbor. A subsequent 3,000-episode diagnostic finds no Holm-significant success difference from the paired anchor (minimum adjusted $p = 0.891$; Appendix A.11). This records conservative proxy-based retention, not evidence that rejecting the proposals prevented task failure or exhausted local improvements.

4.6 HOW MUCH DO OFFLINE PROXIES EXPLAIN?

Initializer signal ablation. At identical static bytes and with the same post-allocation stack, CS+CKA-DiT reaches 53.26% on the Primary-46 slice, versus 51.52% for CKA-DiT alone and 50.65% for CS alone. The +1.74 and +2.61 point differences favor fusion descriptively, but both have Holm $p = 0.5796$ (Appendix A.8). They do not establish a statistically resolved benefit of the fused signal.

Deviation is not success ranking. A separate 60-mask, exact-byte diagnostic tests offline proxy validity across six distances from M_0 . Pooled Spearman correlations with success degradation are -0.116 for D_{PAC} and -0.028 for MSE. Distance confounds this comparison; within-distance D_{PAC} correlations are positive but imprecise, and it does not consistently outperform MSE or D_{func} (Appendix A.15). We therefore interpret the metric as a predeclared deviation guard, not a validated surrogate optimizer for closed-loop success.

5 CONCLUSION AND LIMITATIONS

Quantizing a policy that influences its future inputs requires evaluating more than local tensor fidelity. DYPAC-VLA combines exact-byte precision adaptation with input-conditioned activation quantization, yielding 54.0% GR00T success at $2.22\times$ static compression and 27.7% on $\pi_{0.5}$ at $2.70\times$. Fixed-mask interventions identify deployment range construction as consequential, while the non-monotonic compression sweep cautions against treating FP16 layer count as a proxy for behavioral quality.

The evidence bounds these results. The static-range intervention does not isolate every source of distribution shift. Deviation guards do not predict success reliably, and GR00T retains its initializer without candidate-state adjudication. Its finite candidate set establishes no optimality. Evaluation is in simulation, some baselines require non-equivalent adaptations, and the generic backend saves memory but increases latency and energy. The supported result is a practical success-storage tradeoff whose behavior depends on deployment quantization.

AI USE STATEMENT

Generative AI tools assisted with planning, implementation, debugging, aggregation, writing, formatting, and figure preparation. The authors reviewed all generated material, tested AI-assisted code, reran the evidence and document gates, and verified claims against frozen manifests and primary sources. No experimental result was synthesized by a generative model; the authors take responsibility for the paper and its artifacts.

REPRODUCIBILITY STATEMENT

Sections 3 and 4 specify the method, evaluation, and statistical protocols. The supplement documents content-addressed plans, manifests, coverage checks, table generation, and claim–evidence audits. Headline cells are regenerated from frozen summaries.

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A SUPPLEMENTARY EVIDENCE

A.1 ACTION-WEIGHTED CS-CKA ALLOCATION DETAILS

We construct the GR00T initializer before the full-context search. Its representation-stability score is $S_i = (\lambda_g \bar{d}_i^{\text{geom}} + \lambda_d \bar{d}_i^{\text{dist}}) / (\lambda_g + \lambda_d)$, where $\bar{d}_i^{\text{geom}}$ and $\bar{d}_i^{\text{dist}}$ are jointly normalized versions of $1 - \widehat{\text{CKA}}_i$ and $\widehat{D}_{\text{CS},i}$, respectively. Action-impact weights w_i prioritize layers whose one-layer intervention changes the physical action output, while RMS and saturation guards retain unstable layers in FP16. A greedy byte-constrained solver followed by fixed-size layer flips produces nine complete candidate masks. Paired adjudication on 16 calibration observations selects one mask for each weighting without using success labels. Together, these steps form the guarded representation-stability initializer.

A four-task development sweep then selects $\lambda_g : \lambda_d = 16 : 1$ by closed-loop macro success, yielding the 100-W4/16-FP16 mask used as M_0 . CKA and CS statistics are established tools; the claimed design unit is their constrained conversion into a deployable precision mask followed by the full-context audit, not either statistic in isolation.

The full-context stage runs once from the frozen M_0 ; it does not restart from several initial masks and retain the best-performing one. The artifact registry records the exact initializer and calibration-buffer hash. The four ratio-selection tasks remain in the 50-task aggregate. A post-freeze calculation over the other 46 tasks gives 51.8% for ours, 27.6% for QuantVLA W4A8, and 52.8% for FP16, showing that the four development tasks do not create the main pattern.

A.2 EXACT METRIC SPECIFICATION AND EDGE CASES

Sequence construction and action semantics. Each metric unit is a task-seed sequence u with R_u unique replans. We sort replans by their recorded integer index; each contributes the complete inverse-normalized (16, 12) chunk executed before the next replan, so $N_u = 16R_u$. We discard later unexecuted forecasts, do not overlap adjacent chunks, and do not deduplicate predictions. A sequence contains only complete chunks collected before success, termination, truncation, or the official horizon. Duplicate replans, any shape other than (16, 12), or a non-finite value fail the metric gate rather than being padded or imputed. Coordinates 1–3 encode end-effector translation, 4–6 a rotation vector, 7 gripper close, and 8–12 the deployed base/control commands. Equation 9 normalizes each coordinate separately.

For coordinate k , we estimate a robust scale once from the complete FP16 selection buffer and never recompute it per task, sequence, or candidate:

$$\tilde{s}_k = \max(1.4826 \text{MAD}_k, 0.1 \text{RMS}_k), \quad s_k = \max(\tilde{s}_k, 0.05 \text{median}_j \tilde{s}_j, 10^{-4}). \quad (9)$$

With $z_{u,t} = (a_{u,t}^Q - a_{u,t}^T)/s$ and $\rho(z) = 2(\sqrt{1+z^2} - 1)$, the local and accumulated terms are

$$d_{\text{local}}^{(u)} = \frac{1}{N_u d} \sum_{t,k} \rho(z_{u,t,k}), \quad d_{\text{prefix}}^{(u)} = \frac{1}{N_u d} \sum_{t,k} \rho\left(\frac{\sum_{j=1}^t z_{u,j,k}}{\sqrt{t}}\right). \quad (10)$$

Pose map. For $\xi = [v; \omega]$, let $\Omega = [\omega]_{\times}$, $\theta = \|\omega\|_2$, and

$$\begin{aligned} \text{Exp}(\xi) &= \begin{bmatrix} I + A\Omega + B\Omega^2 & (I + B\Omega + C\Omega^2)v \\ 0 & 1 \end{bmatrix}, \\ A &= \frac{\sin \theta}{\theta}, \quad B = \frac{1 - \cos \theta}{\theta^2}, \quad C = \frac{\theta - \sin \theta}{\theta^3}. \end{aligned} \quad (11)$$

The implementation uses Taylor limits for $\theta < 10^{-5}$. Right multiplication, $T_t = T_{t-1} \text{Exp}(\xi_t)$, interprets each increment in the current end-effector frame. The logarithm uses the principal rotation angle and inverse left Jacobian $V^{-1} = I - \Omega/2 + c\Omega^2$, where $c = 1/12$ near zero and $c =$

$\theta^{-2} - (1 + \cos \theta)/(2\theta \sin \theta)$ otherwise. The three control-specific components are

$$e_t^{\text{pose}} = \frac{\text{Diag}(s_{1:6})^{-1} \text{Log}((T_t^T)^{-1} T_t^Q)}{\sqrt{t}}, \quad d_{\text{pose}}^{(u)} = \bar{\rho}_t(e_t^{\text{pose}}), \quad (12)$$

$$d_{\text{stitch}} = \bar{\rho} \left(\frac{(A_{r+1,1}^Q - A_{r,H}^Q) - (A_{r+1,1}^T - A_{r,H}^T)}{s} \right), \quad (13)$$

$$d_{\text{grip}}^{(u)} = \frac{1}{2} \left[\bar{\rho}_t(g_t^Q - g_t^T) + \bar{\rho}_{t>1}((g_t^Q - g_{t-1}^Q) - (g_t^T - g_{t-1}^T)) \right], \quad (14)$$

where $g_t^P = \sigma(4a_{t,7}^P/s_7)$. Translation and rotation are normalized coordinatewise before averaging.

Functional diagnostic. For physical final chunks indexed by context i , define

$$e_i = \frac{\|A_i^Q - A_i^T\|_F^2}{\max(\|A_i^T\|_F^2, 10^{-8})}, \quad d_{\text{final}} = \frac{\|A^Q - A^T\|_F^2}{\max(\|A^T\|_F^2, 10^{-8})}, \quad (15)$$

$$D_{\text{func}} = d_{\text{final}} + d_{\text{kin}} + d_{\text{grip}} + 2 \text{CVaR}_{0.9}(d_{\text{solver}}).$$

d_{kin} is one half of the sum of analogous relative errors restricted to translation and rotation coordinates. We set $d_{\text{solver}} = \text{mean}_i e_i$, and $\text{CVaR}_{0.9}$ averages e_i at or above its empirical 90th percentile. The registered D_{func} gripper term is the fraction of adjacent solver states for which active gripper changes ($> 10^{-4}$ in both policies) disagree in sign. Because the shared selection path records only final physical chunks and not the internal solver trajectory, this legacy D_{func} gripper term is zero on that path. Gripper deviation still enters D_{PAC} through Equation 14 and the explicit component guard.

Evaluation order. For each candidate, we (1) validate paired records, sort replans, and inverse-normalize final chunks; (2) compute the global 12-coordinate scale once from all FP16 selection chunks; (3) evaluate Equations 10, 12, 13, and 14 for each task-seed sequence; (4) form the five-component mean and worst-10% sequence mean in Equation 4; and (5) compute task-level candidate-minus- M_0 differences, the jackknife standard error in Equation 16, J , and the pose/stitch/gripper upper bounds. We set $d_{\text{stitch}} = 0$ for a one-replan sequence and the gripper-difference term to zero for a one-action sequence. Acceptance requires strict $J_{\text{final}} < 0$, non-positive component upper bounds, and the exact byte constraint; equality fails closed.

For n_g task scalars x_t in cluster g , the implemented leave-one-task-out jackknife standard error is

$$\text{SE}^{\text{jack}}(\bar{x}) = \sqrt{\frac{\sum_{t \in g} (x_t - \bar{x})^2}{n_g(n_g - 1)}}, \quad (16)$$

which is algebraically identical to the usual sample standard error of the task mean.

A.3 FROZEN GR00T DECISION RECORD

The GR00T search starts from frozen M_0 . Corrected task-level statistics identify two of 116 coordinates with positive conservative benefit for retaining FP16 under both D_{func} and D_{PAC} . Both are already protected: removing their protection increases the proxy loss, rather than providing two improving flips. The generator rebuilds six complete protected sets containing one to five FP16 layers, not small edits of the 16-layer anchor. All six are scored on teacher-state contexts over the three registered splits. Their objectives lie in $[0.948, 2.360]$; each also fails at least one component guard. The protocol retains M_0 without candidate-state scoring. This decision neither exhausts the feasible neighborhood nor establishes superiority in task success; no prior rollout, byte count, or checkpoint is changed.

The final plan uses signed-nibble group-64 weights, identity row rotation, and per-forward dynamic A8, with no ATM, OHB, error folding, calibration bypass, or runtime selector. Its SHA-256 is recorded in the artifact registry and checked before table generation.

A.4 MASK DIAGNOSTIC ROBUSTNESS

The registered 50-episode diagnostic compares M_0 with uniform full-W4 under an otherwise identical runtime. M_0 obtains 56% versus 46%, with ten paired wins, five losses, and 35 ties ($p = 0.302$). An

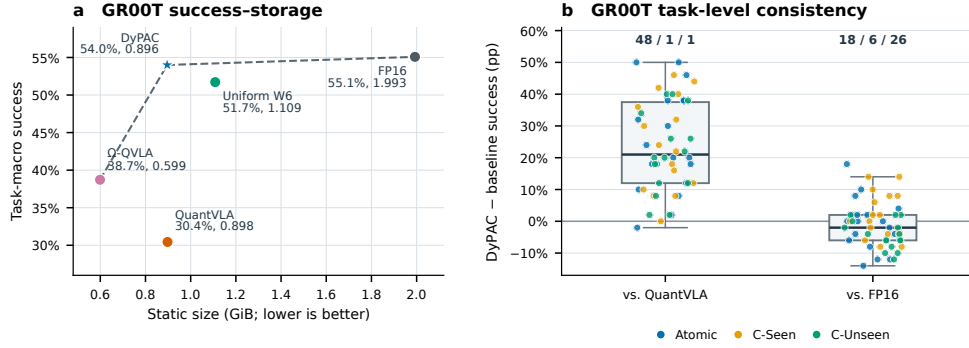


Figure 2: GR00T behavior-storage evidence for selected measured configurations on the 50-task RoboCasa365 target split. **(a)** Task-macro success versus exact static-component storage (2,500 episodes per point). The dashed line connects displayed non-dominated points; it is not an interpolation. **(b)** Per-task differences for DyPAC-VLA on matched tasks and seeds. Boxes show the median and interquartile range; counts denote better/tied/worse tasks. Table 1 reports the complete comparison, including ActQuant and the protocol-matched $\pi_{0.5}$ results.

earlier 50-episode exploratory wave reverses the direction (50% versus 52%); pooling both waves gives 53% versus 49% ($p = 0.557$). We retain this small-sample result as a robustness check outside the headline evaluation.

A.5 PAIRED STATISTICAL DETAIL

Against QuantVLA W4A8, 1,647 of 2,500 episodes are tied, 721 favor DyPAC-VLA, and 132 favor QuantVLA. The exact two-sided McNemar test gives $p = 6.30 \times 10^{-99}$; Holm adjustment over the registered comparison family gives $p = 1.89 \times 10^{-98}$. Against FP16, 1,889 episodes are tied, 292 favor DyPAC-VLA, and 319 favor FP16, giving $p = 0.2929$.

Against the initializer-stage GDSQ-VLA deployment, 1,841 episodes are tied, 370 favor DyPAC-VLA, and 289 favor GDSQ-VLA. The exact McNemar test gives $p = 0.00181$, and Holm adjustment gives $p = 0.00362$. All three split-specific GDSQ-VLA plans use exactly the same 100-W4/16-FP16 decision vector as the final mask. GDSQ-VLA uses the original DuQuant W4 substrate and split-calibrated static A8, whereas the final stack uses Hessian-aware W4 and DYRANGE-A8. The contrast therefore measures their combined post-allocation effect.

A.6 ROBOCASA365 SUCCESS-STORAGE DETAIL

Figure 2 places the GR00T result among selected measured success-storage configurations and reports task-level directions against QUANTVLA and FP16. The dashed line connects non-dominated displayed configurations only and is not an interpolation; Table 1 contains the complete comparison, including ActQuant.

A.7 COMPRESSION-RATE SWEEP

Table 3: Six-arm GR00T compression-rate sweep. Success remains on a 51.4–55.4% plateau while static compression increases from 1.166 $\times$ to 2.556 $\times$.

Target r	W4/FP16	Size (GiB)	Static comp.	All-50 SR (%) $\uparrow$	Atomic	C-Seen	C-Unseen
1.20	30/86	1.709	1.166 $\times$	55.4	74.4	46.3	43.1
1.60	63/53	1.359	1.466 $\times$	52.8	71.7	43.8	40.6
2.00	80/36	1.146	1.739 $\times$	54.6	74.4	41.9	45.0
2.40	91/25	1.007	1.979 $\times$	51.4	71.7	42.5	37.5
2.80	98/18	0.907	2.196 $\times$	54.0	73.9	44.4	41.3
Max (3.56)	116/0	0.779	2.556 $\times$	55.4	71.7	48.1	44.4

Every arm uses the frozen FCP ranking, Hessian-g64 W4, deployment-conditioned A8, four flow steps, all 50 tasks, and seeds 0–9 (500 episodes per arm). All 3,000 expected units are unique and complete. Target r applies to candidate tensors; static compression includes fixed components and packing metadata. The frozen 100-W4/16-FP16 headline policy obtains 54.0% at 2.224 $\times$ and is cited rather than rerun.

The sweep isolates storage pressure while retaining one frozen layer ordering and the same Hessian-g64/dynamic-A8 execution stack. Shrinking the static policy by 2.19 $\times$ across the two endpoints produces no monotonic success degradation: the 2.556 $\times$ all-W4 endpoint reaches 55.4%, tying the 1.166 $\times$ arm and exceeding the frozen 2.224 $\times$ headline rate of 54.0%. The isolated 51.4% trough at 1.979 $\times$ is driven most strongly by Composite-Unseen, which falls to 37.5% and recovers to 44.4% at the all-W4 endpoint. Across this candidate domain, the deployment stack sustains a broad high-success compression plateau.

Formal-scale maximal-rate endpoints. The all-W4 endpoints are additionally evaluated at formal Table-1 scale (50 seeds per task, 2,500 episodes) under the shared four-flow-step protocol and are displayed in Table 1 as DYPAC-VLA (Ours, max rate). On GR00T, the all-W4 rate-family endpoint reaches 52.8% task-macro success (1319/2500; 70.9/42.9/42.3 by split) at 836,960,256 static bytes (2.556 $\times$), versus 54.0% for the audited anchor at 2.22 $\times$; the cross-run paired comparison has 319/288 discordant pairs (exact McNemar $p = 0.223$). On $\pi_{0.5}$, the 180-layer all-W4 profile reaches 26.8% (670/2500; 58.7/14.5/3.3 by split) at 1,242,169,344 static bytes (3.556 $\times$), versus 27.7% for the projected anchor and 26.2% for FP16; the cross-run paired comparison has 182/159 discordant pairs (exact McNemar $p = 0.233$). Both rows are descriptive operating points, not registered tests, and neither revises the frozen headline rows. The aggregate is content-addressed at runs/robocasa365_table1_max_sweep_v1/aggregate.json (SHA256 93bb7b021ca83b073b314c6b18e97f3d7b4c4e327eeb190375d7d5fedb3070bc) and the plans at runs/robocasa365_table1_max_sweep_v1/masks/gr00t_max.plan.json (SHA256 b1fe584404184bf6f5505af5ef40ebaeab759a94c1e0e952dd19b085e387179d) and runs/robocasa365_table1_max_sweep_v1/masks/pi05_max.plan.json (SHA256 8b0aabc15e48970250e605ca689e08bde3c9b73616d2c31fd2fae88d01759dbd).

A.8 CS-CKA SIGNAL ABLATION

Table 4: Exact-byte GR00T signal ablation. Both primary comparisons favor CS+CKA-DiT, but the evidence is directional rather than significant.

Signal	W4/FP16	Primary-46 SR (%) $\uparrow$	Fused Δ (pp)	Holm p	All-50 SR (%) $\uparrow$
CS+CKA-DiT (16:1)	93/23	53.26	–	–	55.20
CKA-DiT only	94/22	51.52	+1.74	0.5796	53.20
CS only	92/24	50.65	+2.61	0.5796	52.40

All arms use exactly 962,068,480 static bytes (2.2239 $\times$ compression), deployment-conditioned A8, 50 tasks, and seeds 0–9, totaling 1,500 episodes. Primary-46 excludes the four tasks used to select the 16:1 ratio. Deltas are fused minus the single-signal arm; the two paired McNemar tests form one Holm family.

This preregistered ablation changes only the allocation signal while holding the exact static byte budget, Hessian-aware W4, deployment-conditioned A8, and rollout protocol fixed. The Primary-46

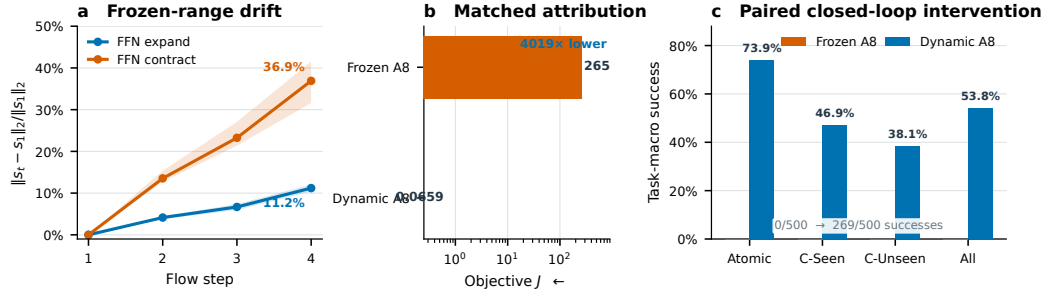


Figure 3: Evidence for per-forward A8 scaling. (a) Calibrated per-channel ranges drift across flow steps. (b) Matched offline attribution relative to A16 under objective J (log scale). (c) Under an otherwise identical M_0 runtime, the registered static table obtains 0/500 successes and dynamic A8 obtains 269/500. The intervention evaluates this specific static construction, not every possible static activation quantizer.

estimand excludes the four tasks used to choose the 16:1 fusion ratio. CS+CKA-DiT is higher than CKA-DiT only on 16 tasks, tied on 17, and lower on 13; against CS only the corresponding counts are 21/11/14. Both paired comparisons point toward fusion, with a common Holm-adjusted $p = 0.5796$, providing directional evidence for the combined signal.

A.9 REDUCED CLOSED-LOOP COMPONENT ABLATION

The earlier reduced component study uses all 50 tasks and ten paired seeds per task. It is disjoint from mask construction and was aggregated only after all 1,500 new episodes passed the coverage audit. The matched M_0 reference obtains 73.9%, 46.9%, and 38.1% on Atomic-Seen, Composite-Seen, and Composite-Unseen, respectively. Its overall success rate is 53.8%.

Table 5: GR00T post-initialization controls. The identical-mask stack gains +3.2 points; the outcome-blinded FCP diagnostic finds no Holm-significant proposal.

Configuration	W4/FP16	Episodes	SR $\uparrow$	Δ (pp)
GDSQ-VLA (DuQuant/static A8)	100/16	2500	50.8	reference
DyPAC-VLA (Hessian/DYRANGE-A8)	100/16	2500	54.0	+3.2
M_0 + DYRANGE-A8	100/16	500	53.8	reference
Frozen-range A8	100/16	500	0.0	-53.8
FCP proposals (6; diagnostic)	111-115/5-1	3000	50.6-56.2	-3.2-+2.4

Candidate success labels were revealed only after FCP froze M_0 . The smallest Holm-adjusted p across the six paired tests is 0.891. Table 7 reports every proposal.

Frozen-range A8 obtains 0.0% on every split. All 500 episodes complete normally, with no crash or numerical-failure row. Relative to M_0 , it has no paired wins, 269 losses, and 231 ties. Under the registered calibration protocol, range construction alone determines the transition from 0 to 269 successful episodes.

Local-MSE selection obtains 72.8%, 43.8%, and 40.0% across the three splits, for 53.0% overall. Against M_0 , it has 55 paired wins, 59 losses, and 386 ties. The no-final-check proposal obtains 70.6%, 38.8%, and 47.5%, also giving 53.0% overall, with 54 wins, 58 losses, and 388 ties. Their different split behavior—especially the seen/unseen reversal for the no-final-check proposal—shows why the full 50-task result is more informative than results on the two seen groups alone. Equal aggregate rates do not imply equivalent policies.

A.10 DYNAMIC-RANGE EVIDENCE

Figure 3 separates mechanism diagnosis from behavioral evidence. Across the 31 applicable action-head layers, median scale drift reaches 10.8%, 16.8%, and 24.9% at flow steps 2-4; the maximum layer-step value is 49.0%. With the corrected task-level standard error, the registered static table

Table 6: Core offline ablations on the frozen 12-task, 36-sequence context buffer; lower objective is better.

Factor	Controlled counterfactual	Audited evidence	Decision
Range	Static A8 vs. FP16-activation control	$J = 264.94$	Reject static A8
Range	Dynamic A8 vs. registered static A8	$J = -0.241$	Use DYRANGE-A8
Proposal audit	FP16-retention value from state flips	2/116 positive retention values	Rebuild protected sets
Policy audit	6 corrected complete policies	0/6 eligible; $J \in [0.95, 2.36]$	Retain M_0

J is the worst normalized one-standard-error upper bound over D_{PAC} , D_{func} , and task clusters, using the corrected jackknife SE of the task mean. Dynamic-vs-A16 has $J = 0.066$. All six corrected candidates have completed teacher-state scoring; none passes the minimax and component guards, so candidate-state scoring is not triggered.

has $J = 264.94$ against the A16 diagnostic. DYRANGE-A8 reduces this discrepancy to $J = 0.066$ and improves over the static table with $J = -0.241$. Under an otherwise identical M_0 runtime, the paired closed-loop intervention gives 0/500 successes for the registered static table and 269/500 for DYRANGE-A8.

A.11 FULL-POLICY AUDIT OUTCOME

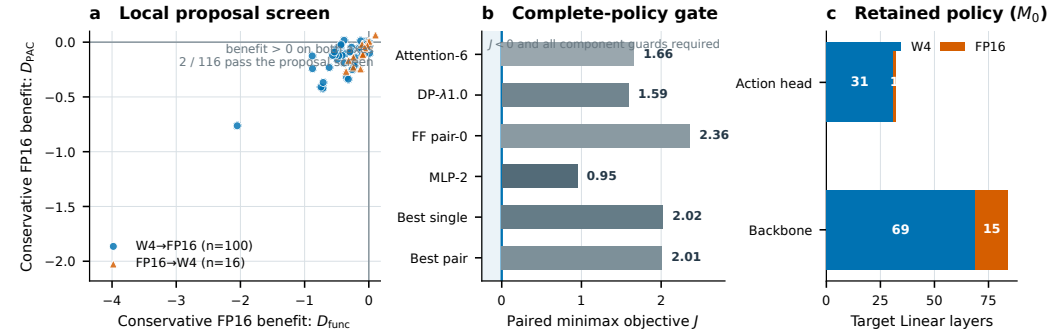


Figure 4: GR00T deviation-based decision record with corrected task-level statistics. (a) Two of 116 coordinates have positive conservative benefit for retaining FP16 under both metrics. Both are already protected; these are not two improving removals. (b) All six rebuilt complete-policy candidates have $J > 0$ and fail a component guard. (c) The 100-W4/16-FP16 initializer is retained. This decision does not establish its success optimality or demonstrate an effect of candidate-state adjudication, which was not triggered.

Figure 4 separates conservative FP16-retention benefit from complete-policy acceptance. Two coordinates have positive retention benefit under both metrics, but neither is an improving FP16-to-W4 flip. All six rebuilt candidate masks have positive minimax objectives ($J = 0.948$ – 2.360) and fail the component-deviation gate. The 100-W4/16-FP16 M_0 remains unchanged. Local retention scores do not establish that removing other protections simultaneously will yield an acceptable configuration.

The candidate set, task–seed coverage, paired tests, and Holm family were frozen before outcomes were revealed, and the preregistration forbids feeding these results back into selection. The largest observed difference is $+2.4$ points for the FF-pair proposal (raw $p = 0.285$, Holm $p = 1.000$); the minimum adjusted value across the family is 0.891. These post-selection outcomes leave task-success differences unresolved; they do not validate rejection as preventing a regression. Outcomes are reported without retrospectively selecting the most favorable rollout estimate.

A.12 REAL-HARDWARE MEASUREMENT DETAIL

Relative to native FP16, M_0 reduces median CUDA peak allocation from 5.386 to 4.267 GiB ($0.792\times$) and median NVML device use from 7,644 to 6,720 MiB. Median latency nevertheless rises from 107.5

Table 7: Preregistered post-selection GR00T diagnostic. None of the six FCP proposals differs from the frozen M_0 after Holm correction.

Configuration	W4/FP16	SR $\uparrow$	Δ (pp)	W/L	Raw p	Holm p
M_0 (frozen)	100/16	53.8	—	—	—	—
Single-layer	115/1	54.8	+1.0	59/54	0.707	1.000
Two-layer	114/2	50.6	-3.2	46/62	0.149	0.891
Attention block 6	112/4	55.6	+1.8	56/47	0.431	1.000
MLP block 2	113/3	53.0	-0.8	53/57	0.775	1.000
FF pair	114/2	56.2	+2.4	59/47	0.285	1.000
DP- λ 1.0	111/5	52.8	-1.0	54/59	0.707	1.000

Each proposal uses 50 tasks and ten seeds per task (500 episodes); Δ and paired wins/losses are relative to the same 500-episode M_0 control. The six tests were fixed before rollout and adjusted as one Holm family.

Table 8: Real GR00T hardware measurements on an otherwise idle NVIDIA A40 at batch size one. The final two rows form a separate paired backend A/B; lower is better for every measured column.

Configuration	W4	p50 (ms)	p95 (ms)	Peak alloc. (GiB)	Gross J/req.	Idle-adj. J/req.
Native FP16	0	107.5	143.2	5.386	18.65 $\pm$ 4.23	10.65
QUANTVLA W4A8	100	269.6	297.2	4.319	40.76 $\pm$ 1.15	21.52
M_0 (ours)	100	232.2	258.3	4.267	39.81 $\pm$ 0.46	22.77
Single-best	115	245.2	271.0	4.165	41.58 $\pm$ 0.24	23.68
Two-best	114	245.8	268.4	4.171	41.40 $\pm$ 0.41	23.42
Attention-6	112	245.5	258.0	4.165	41.32 $\pm$ 0.27	23.60
MLP-2	113	240.6	261.2	4.200	40.45 $\pm$ 0.23	22.76
FF-pair-0	114	244.4	267.5	4.172	41.48 $\pm$ 0.57	23.46
DP- λ 1.0	111	238.2	293.5	4.218	40.94 $\pm$ 0.84	23.17
M_0 paired reference	100	229.7	239.4	4.267	39.25 $\pm$ 0.24	22.65
M_0 real W4A8 integer	100	187.5	199.3	4.267	34.64 $\pm$ 0.22	20.97

Each cell aggregates three trials of 60 measured requests after ten warmups (180 requests/configuration), with four flow steps and CUDA synchronization. p50 is the median of trial medians; p95 is pooled. Peak allocation is the median trial CUDA peak. Board energy is trapezoidal integration of 10-Hz NVML power samples; idle-adjusted values subtract a five-second pre-trial baseline. The paired reference and real-integer row must be compared with each other, not across measurement waves. Measurements exclude model loading, simulator, transport, host energy, and robot energy.

to 232.2 ms (2.16 $\times$), and gross board energy rises from 18.65 $\pm$ 4.23 to 39.81 $\pm$ 0.46 J/request (2.14 $\times$). The six rejected candidates span 238.2–245.8 ms, 4.165–4.218 GiB, and 40.45–41.58 J/request. The packed representation therefore provides a measured memory benefit, but neither static bytes nor W4-layer count predicts speed or energy under this generic nibble-dequantize-then-FP16-GEMM backend.

Real-integer backend revalidation. We next replace the generic execution path with packed W4 residency, dynamic signed A8, and INT8 $\times$ INT8 $\rightarrow$ INT32 GEMM. The runtime-contract audit verifies 4-bit weights, 8-bit activations, integer GEMM, the absence of an FP-weight-sized buffer, 409,468,928 packed-weight bytes, and consistent metadata across all three task splits. Kernel compatibility changes activation scaling from the frozen policy’s per-input-channel rule to per-row group-64. We therefore treat this result as a separately named post-hoc deployment point that cannot reopen FCP.

The complete paired closed-loop rerun contains no crash, missing key, duplicate, or terminal failure. The integer backend records 251/500 successes (50.2%) versus 269/500 (53.8%) for the frozen reference: -3.6 points, with 36/54/410 wins/losses/ties, exact two-sided McNemar $p = 0.07255$, and a 10,000-draw task-cluster bootstrap interval of $[-8.8, +1.6]$ points (seed 20260903). The original fake-quant operating point remains the formal policy result, while this row quantifies the separately named integer deployment backend.

In a separate paired hardware wave, the real-integer backend reduces p50 latency from 229.7 to 187.5 ms (18.4%) and pooled p95 from 239.4 to 199.3 ms. Gross board energy falls from 39.25 $\pm$ 0.24 to 34.64 $\pm$ 0.22 J/request (11.8%), while idle-adjusted energy falls from 22.65 $\pm$ 0.35 to 20.97 $\pm$ 0.19 J/request (7.4%). Median peak CUDA allocation remains 4.267 GiB. Each row uses three

60-request trials after ten warmups on the same otherwise idle A40. These model-only measurements exclude simulator, transport, host, and robot energy.

A.13 STORAGE ACCOUNTING

The GR00T FP16 static-component scope is 2,139,537,408 bytes. The final plan occupies 962,068,480 bytes, including 327,352,320 fixed bytes plus packed mixed-precision target components, for $2.2239\times$ compression. This is $0.9982\times$ the 963,772,416-byte QuantVLA cell and satisfies the preregistered $1.10\times$ ceiling. For $\pi_{0.5}$, the corresponding FP16 scope is 4,416,602,112 bytes, and the 1,634,828,288-byte plan gives $2.7016\times$ compression.

These values are static packed estimates. Activations, CUDA workspaces, simulator state, and eager fake-quant caches are outside the scope, so the compression ratios alone imply no runtime benefit. Table 8 instead measures model-only latency, CUDA and NVML peak memory, and GPU-board energy directly on one A40, excluding simulator, transport, host, and robot energy.

A.14 $\pi_{0.5}$ FCP DIAGNOSTIC AND FORMAL RESULT

The transferred $\pi_{0.5}$ initializer contains 80 W4/100 FP16 layers and occupies 1,845,100,544 bytes, exceeding the 1,639,513,497-byte ceiling. FCP evaluates all 180 one-layer precision-state interventions in complete-policy context, then converts the 41 least damaging FP16 protections to W4. The resulting 121-W4/59-FP16 anchor occupies 1,634,828,288 bytes. Subsequent exact-byte proposals do not clear the minimax and component guards, so the projected anchor becomes the final policy.

Table 9: Primary cross-architecture FCP result on $\pi_{0.5}$: complete-policy projection changes 41 precision decisions and makes the transferred initializer byte-feasible.

Configuration	W4/FP16	Size (GiB)	SR $\uparrow$	Δ (pp)	Paired p	Status
M_{init} (transferred)	80/100	1.718	26.0	−2.2	0.235	Over budget
Projected M_0	121/59	1.523	28.2	reference	–	Deployed
Single-best proposal	179/1	1.202	24.8	−3.4	0.086	Rejected
Two-best proposal	178/2	1.247	25.0	−3.2	0.086	Rejected

Each configuration uses the same 50 tasks and seeds 0–9 (500 paired episodes). Deltas are relative to projected M_0 . The projection comparison is the preregistered singleton primary test ($p = 0.235$); the two proposal values are Holm-adjusted. The frozen choice is projected M_0 .

A separately preregistered, same-protocol diagnostic evaluates all four masks on the same 50 tasks and seeds 0–9. The projected anchor records 141/500 successes versus 130/500 for the transferred initializer, with 41/30/429 paired wins/losses/ties and exact two-sided McNemar $p = 0.235$. Thus the projection meets the exact byte constraint while the observed success difference favors the projected mask; the test establishes neither equivalence nor non-inferiority. The single- and two-best proposals record 124/500 and 125/500 successes; both are lower than the anchor after the frozen selection.

The frozen policy uses four flow steps on the complete RoboCasa365 target split, matching the completed reference rows, with 50 seeds on each of 50 tasks. It completes all 2,500 episodes with no missing, duplicate, or failed row. The policy records 536/900 Atomic-Seen, 123/800 Composite-Seen, and 34/800 Composite-Unseen successes, for 693/2,500 overall; the corresponding task-macro rates are 59.6%, 15.4%, 4.3%, and 27.7%. This establishes the paper’s primary success–storage operating point at $2.702\times$ compression.

A.15 EXACT-BUDGET OFFLINE PROXY PREDICTIVE VALIDITY

As an auxiliary diagnostic, we test whether frozen offline scores rank the closed-loop consequences of precision allocation. The library contains 60 masks with identical 100-W4/16-FP16 allocation and exact static bytes, stratified into ten masks at each of six swap distances from M_0 . We freeze offline scores before running one rollout per target task and mask. Across all masks, Spearman correlation with signed success degradation is -0.116 for D_{PAC} and -0.028 for MSE. Mask distance confounds this pooled comparison: larger swaps have greater offline distortion but, in this library, slightly higher mean success. Within each fixed-distance stratum, the registered D_{PAC} correlations are positive but

imprecise ($\rho = 0.055\text{--}0.352$, ten masks per stratum), and D_{PAC} does not consistently outperform MSE or D_{func} . These results position D_{PAC} as a predeclared complete-policy deviation guard rather than a success-ranking objective.

A.16 ACTQUANT LOCAL CROSS-ARCHITECTURE ADAPTATION

We pin the official ActQuant implementation at commit `b64791125070652fe6b554e244fe809c79ef5246` and retain its allocation and quantization semantics Akbari et al. (2026). Because local adapters provide the GR00T, $\pi_{0.5}$, and RoboCasa365 integration, we label the result `source_protocol_equivalent=false`. Calibration uses an FP16-teacher proxy and excludes formal seeds 0–49; no formal outcome selects or tunes the packs.

Both ActQuant configurations complete all 50 tasks and seeds 0–49 under the four-flow-step protocol. GR00T records 649/900, 341/800, and 332/800 successes on Atomic-Seen, Composite-Seen, and Composite-Unseen, respectively, for 1,322/2,500 (52.9%) overall. The corresponding $\pi_{0.5}$ counts are 509/900, 98/800, and 22/800, for 629/2,500 (25.2%). Against DYPAC-VLA, ActQuant has 340/368 paired wins/losses on GR00T (Holm $p = 0.3102$) and 135/199 on $\pi_{0.5}$ (Holm $p = 0.0011$). Its measured static packs occupy 3.655 GiB and 2.904 GiB, respectively; these sizes do not imply runtime-memory or latency gains.

A.17 QVLA-CODE PUBLIC-RELEASE REPRODUCTION FAILURE

We cloned the official QVLA GitHub repository at commit `26cc4821a3be4c003d09d3c7997b38db2a347982` and followed its released Hessian proxy, channel-gate set $\{0, 2, 4, 8, 16\}$, greedy allocation rule, and row-wise fake-quantization semantics Xu et al. (2026b). The release provides QVLA quantization and evaluation paths for OpenVLA-family models on LIBERO, but not GR00T N1.5 or $\pi_{0.5}$ tensor mappings, a deployed mixed-row exporter for these architectures, or a RoboCasa365 evaluation path. We therefore add local cross-architecture adapters and label the run `source_protocol_equivalent=false`.

Under the shared four-flow-step RoboCasa365 protocol, the content-bound run produces 0 successes in 1,728 completed GR00T episodes and 0 successes in 1,517 completed $\pi_{0.5}$ episodes, despite passing local source-parity, pack-loading, and runtime-metadata checks. We stop the run and omit QVLA-code from Table 1. Because the public repository lacks the target-architecture and target-benchmark paths, the outcome is recorded as a failed cross-architecture reproduction; the native OpenVLA/LIBERO result is unaffected.

A.18 DA-PTQ SOURCE REFERENCE AND LOCAL ADAPTATION

DA-PTQ reports CogACT W4A8 mixed-precision experiments calibrated with 512 BridgeData V2 trajectories, together with 42.5% memory reduction and 54.8% speedup in its WidowX setup. These measurements use a different model, benchmark, calibration set, hardware path, and accounting scope from our eager fake-quant RoboCasa365 experiments. Table 10 therefore presents them only as source-reported context, without numerically ranking them against our local results.

Our frozen clean-room adaptation follows the paper-level allocation and transformation semantics while adding GR00T, $\pi_{0.5}$, and RoboCasa365 mappings absent from the public release. Both local configurations complete all 50 tasks and seeds 0–49 under the four-flow-step protocol. GR00T records 73/900, 0/800, and 3/800 successes on Atomic-Seen, Composite-Seen, and Composite-Unseen, respectively, for 76/2,500 (3.0%) overall. The corresponding $\pi_{0.5}$ counts are 527/900, 98/800, and 30/800, for 655/2,500 (26.2%). Against DYPAC-VLA, DA-PTQ has 8/1,282 paired wins/losses on GR00T and 182/220 on $\pi_{0.5}$; the latter comparison has Holm $p = 0.0648$. We report these results as a complete local cross-architecture adaptation with `source_protocol_equivalent=false`, not as a reproduction of the native CogACT/SimplerEnv results.

Table 10: Source-reported task-average success rates (%) from DA-PTQ [Xu et al. \(2026a\)](#). These CogACT/SimplerEnv values are not protocol-matched to our RoboCasa365 results.

SimplerEnv setting	CogACT FP	VLA-Cache	QuantVLA	DA-PTQ
WidowX visual matching	51.3	46.8	43.5	48.9
Google Robot visual matching	74.8	74.4	65.6	68.5
Google Robot variant aggregation	61.3	62.3	49.4	51.7

Values are transcribed from Tables 2–3 of the cited paper. Different models, task distributions, calibration, and evaluation protocols preclude a numerical ranking against Table 1.

A.19 LIBERO PROTOCOL DETAIL

Table 11: LIBERO: DyPAC-VLA averages 97.3% on $\pi_{0.5}$ and 93.8% on GR00T.

Configuration	Goal $\uparrow$	Spatial $\uparrow$	Object $\uparrow$	Long $\uparrow$	Avg. $\uparrow$	Size (GiB) $\downarrow$
$\pi_{0.5}$						
FP16	95.0	100.0	98.0	94.0	96.8	4.113
QUANTVLA W4A8	97.0	97.0	98.0	93.0	96.3	1.388
Uniform W6	99.0	97.0	99.0	92.0	96.8	1.902
Ω -QVLA W4A4	100.0	99.0	97.0	96.0	98.0	1.307
QVLA (4.0 BPW) [†]	96.4	98.0	97.2	91.8	95.8	2.7
ActQuant (4.0 BPW) [†]	96.8	98.4	99.4	91.8	96.6	2.7
DyPAC-VLA (Ours)	99.0	100.0	96.0	94.0	97.3	1.788

Table 11: LIBERO results (continued): GR00T N1.5.

Configuration	Goal $\uparrow$	Spatial $\uparrow$	Object $\uparrow$	Long $\uparrow$	Avg. $\uparrow$	Size (GiB) $\downarrow$
GR00T N1.5						
FP16	88.0	87.0	96.0	76.0	86.8	1.993
QUANTVLA W4A8	89.0	85.0	79.0	72.0	81.3	0.898
Uniform W6	85.0	83.0	92.0	22.0	70.5	1.109
Ω -QVLA W4A4	87.0	86.0	89.0	76.0	84.5	0.599
DyPAC-VLA (Ours)	97.0	95.0	98.0	85.0	93.8	0.948

The complete local matrix contains 4,000 episodes across 40/40 exact-valid cells. Each local suite uses ten tasks and ten held-out initial states. Size includes packed weights and metadata and is averaged over suite-specific masks. [†]The $\pi_{0.5}$ QVLA and ActQuant rows are source-reported 4.0-BPW results [Akbari et al. \(2026\)](#) using the source memory definition.

Table 11 reports the complete suite matrix. Each local cell contains 100 episodes over ten tasks and held-out states 10–19, and all 40 cells pass the joint gate. The corrected GR00T rerun uses five actions per replan and paired suite-keyed noise; we do not interpret the correction as an algorithmic gain. Local size is the average packed weight and metadata footprint across suite-specific masks. The two source-reported 4.0-BPW $\pi_{0.5}$ anchors [Akbari et al. \(2026\)](#); [Xu et al. \(2026b\)](#) are excluded from the 4,000-episode gate, and their 2.7-GB memory measure is not directly compared with our audited static size.